

## 2.3 One Quantitative Variable: Using Percentiles

- Goal: We need to turn “raw” data into information.
- Data are summarized into
  - 1 tables & graphs
  - 2 numerical indices
- One quantitative variable
  - 1 Five-number summary
  - 2 Percentiles

# Three Features of Data

## ■ What look like the data?

- 1 The **shape** of the distribution of the data.
- 2 While there are many different variety of shapes, certain shapes are common enough to have been named.

## ■ Where is the center of the data?

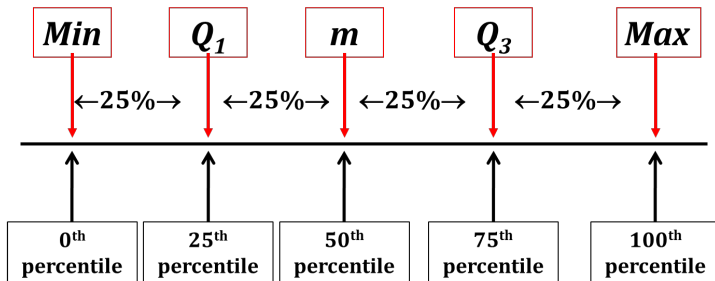
- 1 The **central location** of the distribution of the data is the most important aspect to notice and describe.
- 2 Measures of central location: **Mean**, **Median**, **Mode**.

## ■ How spread out the data?

- 1 The **variability** of the distribution of the data is the second most important feature.
- 2 Measures of spread: **Standard deviation**, **IQR**, **Range**.

# Percentiles and Quartiles

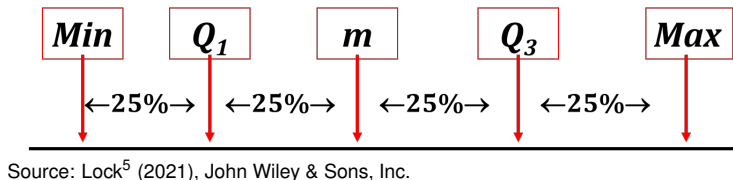
- The  **$P$ th percentile** is the value that, after the data are arranged in the **ascending** order, has  **$P\%$  of the data below** it and  **$(100 - P)\%$  above** it.
  - 1 **25th percentile** is also called the **first quartile** (or **lower quartile**) and denoted by  $Q_1$ .
  - 2 **50th percentile** is also called the **second quartile** (or **middle quartile**) and denoted by  $Q_2$ , which is also the **median  $m$** .
  - 3 **75th percentile** is also called the **third quartile** (or **upper quartile**) and denoted by  $Q_3$ .



Source: Lock<sup>5</sup> (2021), John Wiley & Sons, Inc.

# Finding Quartiles

- A “shortcut” method for finding lower and upper quartiles:
  - 1 Rearrange the data from the smallest to the largest.
  - 2 Divide the rearranged data into two subsets using the median  $m$  as the dividing value.
  - 3  $Q_1$  = the median of the values below the median  $m$ .
  - 4  $Q_3$  = the median of the values above the median  $m$ .



- There are several different ways of computing percentiles and quartiles.

## IQR and Range | Measure of Spread

- The **interquartile range (IQR)** is the distance between  $Q_1$  and  $Q_3$ :

$$\text{IQR} = Q_3 - Q_1$$

- The **range** is the largest value minus the smallest value:

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

- Notes

- 1 IQR is the **range** of the **middle 50%** data values.
- 2 IQR is **less sensitive to outliers** than any other measures of variability because it completely ignores the extremes in the data.
- 3 IQR can be very misleading when the data set is highly concentrated about the median. For instance, the data sets A and E.

## Example 2.12 | IQR

- Compute three quartiles and IQR for the following data set.

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 2 | 3 | 2 | 6 | 4 | 1 | 5 | 6 | 1 |
|---|---|---|---|---|---|---|---|---|

- 1 To find three quartiles, we first arrange the data values in ascending order as follows:

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 2 | 2 | 3 | 4 | 5 | 6 | 6 |
|---|---|---|---|---|---|---|---|---|

- 2 The median is  $m = 3$ , and this median divides the data into the smaller half (1, 1, 2, 2) and the larger half (4, 5, 6, 6).
- 3 We find  $Q_1 = 1.5$ , which is the median of the smaller subset (1, 1, 2, 2).
- 4 We obtain  $Q_3 = 5.5$ , the median of the larger subset (4, 5, 6, 6).
- 5 Therefore, we compute

$$\text{IQR} = Q_3 - Q_1 = 5.5 - 1.5 = 4.0$$

## Exercise 2.6 | IQR

- Consider the following data sets:

|            |    |    |    |    |    |    |                        |
|------------|----|----|----|----|----|----|------------------------|
| Data set A | 10 | 10 | 10 | 10 | 10 | 10 | $\bar{x} = 10, m = 10$ |
| Data set B | 5  | 5  | 5  | 15 | 15 | 15 | $\bar{x} = 10, m = 10$ |
| Data set C | 0  | 0  | 0  | 20 | 20 | 20 | $\bar{x} = 10, m = 10$ |
| Data set D | 7  | 8  | 9  | 11 | 12 | 13 | $\bar{x} = 10, m = 10$ |
| Data set E | 0  | 10 | 10 | 10 | 10 | 20 | $\bar{x} = 10, m = 10$ |

Compute the IQRs for the five data sets.

| Data set | $Q_1$ | $Q_3$ | IQR | Rank |
|----------|-------|-------|-----|------|
| A        |       |       |     |      |
| B        |       |       |     |      |
| C        |       |       |     |      |
| D        |       |       |     |      |
| E        |       |       |     |      |

# Five-number Summary

- The **five-number summary** is composed of
  - 1 Min (the smallest value)
  - 2  $Q_1$  (the first quartile)
  - 3  $m$  (the median)
  - 4  $Q_3$  (the third quartile)
  - 5 Max (the largest value)

## Example 2.12 (revisited) | Five-number summary

- Find the five-number summary for the following data set.

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 2 | 3 | 2 | 6 | 4 | 1 | 5 | 6 | 1 |
|---|---|---|---|---|---|---|---|---|

- To find three quartiles, we first arrange the data values in ascending order as follows:

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 2 | 2 | 3 | 4 | 5 | 6 | 6 |
|---|---|---|---|---|---|---|---|---|

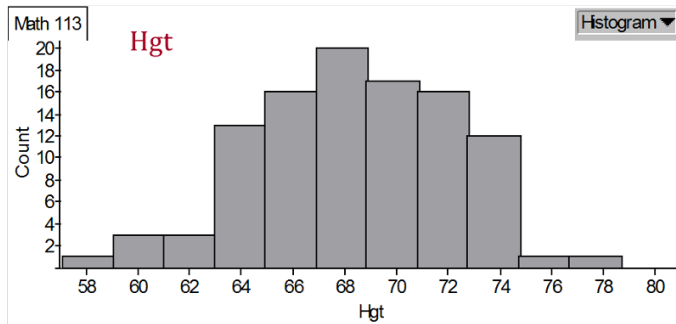
- The median is  $m = 3$ , and this median divides the data into the smaller half (1, 1, 2, 2) and the larger half (4, 5, 6, 6).
- We find  $Q_1 = 1.5$ , which is the median of the smaller subset (1, 1, 2, 2).
- We obtain  $Q_3 = 5.5$ , the median of the larger subset (4, 5, 6, 6).
- The five-number summary is

| Min | $Q_1$ | $\tilde{x}$ | $Q_3$ | Max |
|-----|-------|-------------|-------|-----|
| 1   | 1.5   | 3           | 5.5   | 6   |

## Exercise 2.7 | Five-number summary

■ Based on this histogram of heights, estimate the following:

- 1 The mean
- 2 The standard deviation
- 3 The five-number summary
- 4 The 10th percentile



Source: Lock<sup>5</sup> (2021), John Wiley & Sons, Inc.

# Boxplots

- A **boxplot** (or **box-and-whisker plot**) provides a five-number summary of the data set.
- Procedures for constructing a boxplot:
  - ➊ Mark off a box from  $Q_1$  to  $Q_3$ .
  - ➋ Draw a solid line across the box to locate the median.
  - ➌ Mark the location of the smallest value and the largest value.
  - ➍ Draw a solid line (called “whisker”) between each quartile and the smallest/largest value.
- Example: Draw a boxplot with the following five-number summary.

| Min | $Q_1$ | $\tilde{x}$ | $Q_3$ | Max |
|-----|-------|-------------|-------|-----|
| 1   | 1.5   | 3           | 5.5   | 6   |

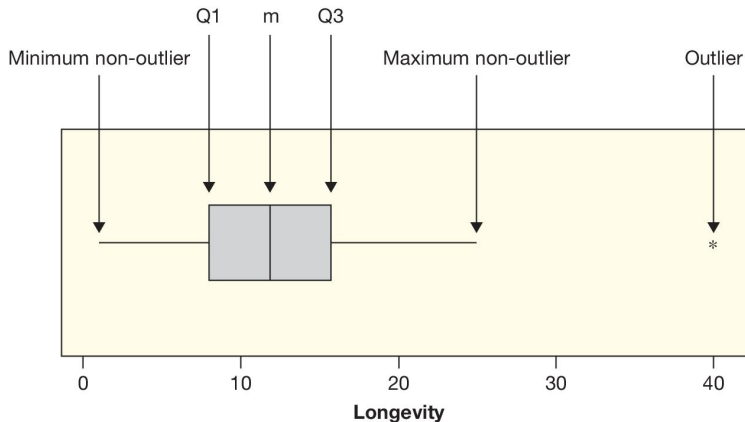
# Detection of Outliers

- The **lower fence** and **upper fence** are defined as follows:
  - 1 Lower fence =  $Q_1 - 1.5 \times \text{IQR}$
  - 2 Upper fence =  $Q_3 + 1.5 \times \text{IQR}$
- **1.5×IQR Rule:** Any observations beyond the fence on either side are called **suspected outliers**.
- Example: Does the  $1.5 \times \text{IQR}$  rule flag any values in the following data set as suspected outliers?

|   |   |   |   |   |   |   |    |   |
|---|---|---|---|---|---|---|----|---|
| 2 | 3 | 2 | 6 | 4 | 1 | 5 | 13 | 1 |
|---|---|---|---|---|---|---|----|---|

## Boxplots – A nicer version

- We can incorporate the information on outliers to the boxplot.



Source: Lock<sup>5</sup> (2026), John Wiley & Sons, Inc.

# Summary | One Quantitative Variable

- Visualizing one quantitative variable:

- 1 Dotplot
- 2 Frequency (or relative frequency) table
- 3 Histogram
- 4 Boxplot

- Shape: symmetric, bell-shaped, skewed

- Summary Statistics

- 1 Measures of center: mean (not resistant to outliers), median (resistant to outliers)
- 2 Measures of spread: standard deviation (not resistant to outliers), IQR (resistant to outliers), range (not resistant to outliers)
- 3 Percentiles
- 4 Five-number summary: min,  $Q_1$ , median,  $Q_3$ , max

- Other concepts

- 1 Outliers, resistance
- 2 z-scores